

Exercises: Simple Gaussian Linear Regression Part III

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(Referring to the theoretical parts: 4, 5, 6, 7, 8)

1 Computer repair data

A corporation sells computer parts and performs maintenance and repair service. The data below have been collected from 18 recent calls to perform maintenance service; for each call, x is the number of repairs and y is the total number of minutes spent by the service person. Assume that a Gaussian linear model $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$ is appropriate. Knowing that:

$$\begin{aligned} \sum_{i=1}^{18} y_i &= 1152 & \sum_{i=1}^{18} x_i &= 81 \\ \sum_{i=1}^{18} (y_i - \bar{y})^2 &= 16504 & \sum_{i=1}^{18} (x_i - \bar{x})^2 &= 74.5 & \sum_{i=1}^{18} (y_i - \bar{y})(x_i - \bar{x}) &= 1098 \end{aligned}$$

where $\bar{x}$ and $\bar{y}$ are the sample means, and

$$\sum_{i=1}^{18} (y_i - \hat{y}_i)^2 = 321.39$$

- Obtain the estimated regression function.
- Interpret β_2 in your estimated regression function. Does β_1 provide any relevant information here? Explain.
- Perform a statistical test with a system of hypothesis $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 > 0$. What can we conclude at 5% significance level?
- Compute a 95% confidence interval for β_2 .
- Test the hypothesis $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 \neq 0$ at a significance level 0.05. Was it necessary to perform this test, given what you already knew from the previous points? Explain.

- f) Obtain a point estimate of the mean service time when $x = 5$ parts are repaired.
Obtain a confidence interval with level 0.90.
- g) Obtain the partition of the total sum of squares and coefficient R^2 .

ASSUMPTIONS

$$n = 18 \quad i = 1, \dots, n$$

$$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i$$

$$\bullet E(\epsilon_i) = 0 \quad \forall i = 1, \dots, n$$

$$\bullet \text{VAR}(\epsilon_i) = \sigma^2 > 0 \quad \forall i$$

$$\bullet \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

$$\bullet \text{COV}(\epsilon_i; \epsilon_k) = 0 \text{ FOR } i \neq k \rightarrow \text{COV}(Y_i; Y_k) = 0$$

$$\bullet X: \text{FIXED AND NON-STOCHASTIC}$$

$$\bullet X: \text{CAN NOT BE ALL EQUAL}$$

$$\bullet \hat{\beta}_2 = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{n-1} \sum (y_i - \bar{y})(x_i - \bar{x})}{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \frac{1098}{74.5} = 14.738$$

$$\bullet \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = \frac{1152}{18} - 14.738 \cdot \frac{81}{18} = -2.321$$

$$Y_i = \beta_1 + \beta_2 x_i = -2.321 + 14.738 x_i$$

- ⑥ (WHEN KEEPING EVERYTHING CONSTANT) ON AVERAGE,
IF WE INCREASE THE NUMBER OF REPAIRS BY 1 UNIT
WE INCREASE THE TOTAL NUMBER OF MINUTES SPENT
BY 14 MIN

$$\bullet \beta_1 \text{ NON USEFUL SINCE NEGATIVE}$$

⑦ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 > 0 \end{cases} \quad \epsilon_{0.05}^{16} \sim \epsilon_{n-p}^{16}$

$$t_{0.05} = \frac{\hat{\beta}_2 - \beta_2}{\text{SE}(\hat{\beta}_2)} = \frac{14.738 - 0}{\sqrt{0.2718}} = 28.26$$

$$\alpha^{0.05} = P(\epsilon_{n-p} > |t_{0.05}|) \approx 1 - 0.9998 \approx 0$$

SINCE $\alpha^{0.05} < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$
I REJECT H_0

$$\text{VAR}(\hat{\beta}_2) = \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} = \frac{\left(\frac{81}{18}\right)^2}{74.5} = 0.272$$

⑧ $1 - \alpha = 0.95 \rightarrow \alpha = 0.05 \rightarrow \alpha/2 = 0.025 \rightarrow 1 - \alpha/2 = 0.975$

$$\epsilon_{n-p; 1-\alpha/2}^{16} = 2.1199$$

$$CI = \left[\hat{\beta}_2 - \text{SE}(\hat{\beta}_2) \epsilon_{n-p; 1-\alpha/2} ; \hat{\beta}_2 + \text{SE}(\hat{\beta}_2) \epsilon_{n-p; 1-\alpha/2} \right]$$

$$[13.632 ; 15.843]$$

- ⑨ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \alpha = 0.05 \quad \text{CAN SKIP WE ALREADY DID} \quad \begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \text{WE WILL REJECT } H_0 \text{ AGAIN}$

③

$$x = 5 \quad \hat{y}_i = -2.321 + 14.738x_i \rightarrow 71.369$$

$$1 - \alpha = 0.90 \rightarrow \alpha = 0.1 \rightarrow \alpha/2 = 0.05 \rightarrow t_{\alpha/2} = 0.95$$

$$y^i \sim N\left(\beta_1 + \beta_2 x^i, \frac{\sigma^2}{n} + \frac{(x^i - \bar{x})^2}{\sum (x_i - \bar{x})^2} \sigma^2\right)$$

$$CI \left(\hat{y}^i - t_{n-p, 1-\alpha/2} \sqrt{s^2 \frac{(x^i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}; \hat{y}^i + t_{n-p, 1-\alpha/2} \sqrt{s^2 \frac{(x^i - \bar{x})^2}{\sum (x_i - \bar{x})^2}} \right)$$

④

$$R^2 = \frac{SSR}{SST} = 0.98$$

$$SST = \sum (y_i - \bar{y})^2 = 16509$$

$$SSE = \sum (y_i - \hat{y}_i)^2 = 321.39$$

$$SSR = SST - SSE = 16187.61$$